decrease until they reach a minimum of about 0.1 solar mass at a circumference of 1000 kilometers and a central density of 3×10^{13} grams per cubic centimeter. This is the first of the neutron stars; it is the "neutron core" that Oppenheimer and Serber studied and showed cannot possibly be as light as the 0.001 solar mass that Landau wanted for a core inside the Sun.

Box 5.6

Unstable Inhabitants of the Gap between White Dwarfs and Neutron Stars

Along the equilibrium curve in Figure 5.5, all the stars between the white dwarfs and the neutron stars are unstable. An example is the star with central density 10^{13} grams per cubic centimeter, whose mass and circumference are those of the point in Figure 5.5 marked 10^{13}. At the 10^{13} point this star is in equilibrium; its gravity and pressure balance each other perfectly. However, the star is as unstable as a pencil standing on its tip.

If some tiny random force (for example, the fall of interstellar gas onto the star) squeezes the star ever so slightly, that is, reduces its circumference so it moves leftward a bit in Figure 5.5 into the white region, then the star's gravity will begin to overwhelm its pressure and will pull the star into an implosion; as the star implodes, it will move strongly leftward through Figure 5.5 until it crosses the neutron-star curve into the shaded region; there its neutron pressure will skyrocket, halt the implosion, and push the star's surface back outward until the star settles down into a neutron-star grave, on the neutron-star curve.

By contrast, if, when the star is at the 10^{13} point, instead of being squeezed inward by a tiny random force, its surface gets pushed outward a bit (for example, by a random increase in the erratic motions of some of its neutrons), then it will enter the shaded region where pressure overwhelms gravity; the star's pressure will then make its surface explode on outward across the white-dwarf curve and into the white region of the figure; and